

Multilingual Generation from Universal Semantic Representation of Sanskrit Text

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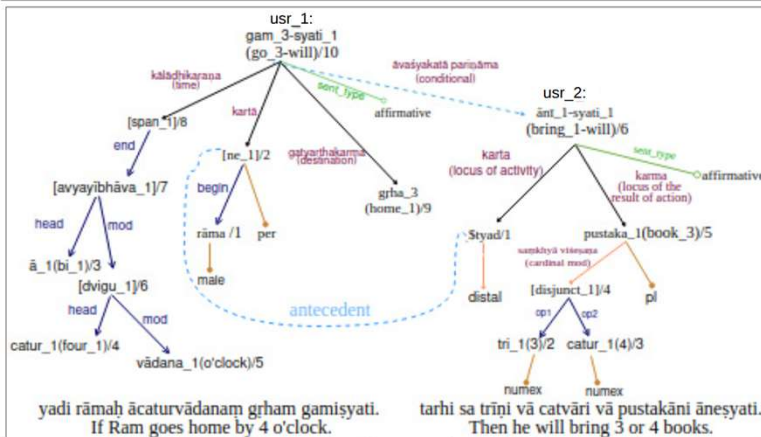
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INTRODUCTION

- Linguistic Challenge: Structural Complexity in Sanskrit
 - complex word formations (sandhi and samāsa), free word order
- Universal Semantic Representation (USR)
 - Language agnostic annotation framework
 - inspired by Paninian and Indian grammatical tradition knowledge
 - captures semantics at: lexical, sentential, discourse, lexico-conceptual, and morphological levels
- Contributions:
 - Propose USR based NLG framework
 - Propose JSON creation from USR as Knowledge Graph input for LLM
 - Experiments highlight advantages of USR as a structured text annotation framework
 - facilitates LLM generate English and Hindi text without training or fine-tuning with Sanskrit language-specific annotated text
 - captures semantics of discourse connectives, discourse particles and pronominal resolution
 - enables LLM-based generators preserve discourse-level information
 - normalize the fluid, synthetic nature of Sanskrit into a rigid, machine-readable format

Proposed NLG Framework

- Strategic shift from raw text to structured Knowledge Graphs (KG) as LLM input reduces hallucinations by providing factual grounding
- Convert USR into JSON-based Knowledge Graph
 - converts positional USR data into explicit triplets (head, relation, tail), mirroring the "attention" mechanism's needs but constraining them with verified semantic data
- By transforming data into triplets, the framework provides the model with a clear logical path, resulting in the high-performance metrics observed in benchmarks
- Proposed custom JSON format:
 - usr_id**: Unique identifier for the semantic unit.
 - SENT_TYPE**: Identifies the sentence as affirmative, interrogative, etc.
 - nodes**: Individual concepts and their properties (e.g., attr_gen, tam).
 - edges_dep**: Dependency triplets capturing the kāraka structure.
 - edges_cxn**: Captures complex concepts like compounds.
 - edges_discourse**: Maps inter-sentential relations like co-reference and logical flow.
- supply JSON format input to LLM with prompts containing instructions to generate output text
- NLG pipeline:
 - USR → JSON → Prompt LLM → output text



Model	Target Language	Human Evaluation Score
Gemini 2.5 Pro	English	0.974
Gemini 2.5 Pro	Hindi	0.982
Gemini 2.5 Flash	English	0.959
Gemini 2.5 Flash	Hindi	0.737
Google Translate	English	0.878
Google Translate	Hindi	0.957

Concepts	Index	Semantic Category	Morpho-Semantic Category	Dependency Relation	Discourse	Speaker's View	Scope	Construction
rāma(Ram,rāma)	1	male	-	-	-	-	-	2:begin
[ne_1]	2	per	-	-	-	-	-	-
sahara_1(city_6,nagara_1)	3	-	-	6:k1	-	def	-	-
ghūma_15(visit_6,bhram_5)	4	-	-	6:k2p	-	-	-	-
nahim_1(not_1,na_1)	5	-	-	6:rt	-	-	-	-
jā_1(go_1,gam_9)-vā_1(ed_1,at_1)	6	-	-	0:main	-	-	-	-
%negative								
</segment_id=1a>								
<segment_id=1b>								
Styad	1	-	-	2:r6	1a.2:coref	distal	-	-
mām_1(mother_8,mātr_1)	2	female	-	4:k1	-	respect	-	-
bimāra_1(sick_1,rūga_2)	3	-	-	4:k1s	-	-	-	-
hai_1(state_copula_1,as_86,ati_4)-past_past,ati_4	4	-	-	0:main	1a.6:parināma	-	-	-
%affirmative								
</segment_id>								
<segment_id=1c>								
Styad	1	-	-	5:k1	1a.2:coref	distal/hī_1	-	-
Styad	2	-	-	5:k2	1b.2:coref	distal	-	-
dekhabhāla_1(care_8,aveks_1)	3	-	-	-	-	-	-	5:kriyāmūla
kara_30(take_15,-)nā_1(had_to_1,et_1)	4	-	-	-	-	-	-	5:verbalizer
[ep_1]	5	-	-	0:main	1b.4:sammuccaya	-	-	-
%affirmative								
</segment_id>								

Challenge	Definition	Impact on MT Accuracy
Sandhi	Phonological modifications at morph or word boundaries.	Severely complicates tokenization and lemmatization by obscuring discrete word boundaries.
Samāsa	Compounding where multiple roots merge into a unified semantic unit.	Requires complex sub-lexical semantic decomposition that standard parsers often fail to execute.
Free Word Order	Flexibility in word arrangement driven by case-inflections rather than syntax.	Renders the "proximity rule" invalid, leading to systemic failures in identifying non-linear modifier-modified relationships.

EXPERIMENTS & RESULTS

- Baseline: Google Translate,
- Human Evaluation done for Google translate and our NLG framework generated outputs
 - Scores improved by 10% for English and 3% for Hindi for our NLG framework
- Human evaluation Parameters:
 - Fact alignment: Is the full information of the input present in the output
 - Relevance: Is more information present in the output than in the input
 - Misinformation: Is wrong information present in the output than the input
 - Coherence: Is the output text coherent, having a good flow of the discourse
 - Grammatical Errors
 - Spelling Mistakes
 - Native like fluency
 - Non-legitimate words of the language

EXPERIMENTS & RESULTS contd..

- For each parameter evaluators provide 0-4 scale rating
 - 0: Poor
 - 1: Less than average
 - 2: Average
 - 3: More than average
 - 4: Excellent
- To obtain **human evaluation score** for a particular model for a specific target language using Min-Max normalization process rating values:
 - For each file, for each evaluator, normalize all parameter scores and obtain average
 - For each file, obtain average scores from all evaluators
 - Obtain average scores from all files as **average normalized human evaluation score** of a model for a target language